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Add-ons, Firefox's most touted feature, have finally come to Google Chrome, and you'll be surprised to know how easy it is to get started making one. While Chrome extension support is certainly not as powerful as that of Firefox, it's much simpler and easier to use.

Chrome's simple extension support comes with many advantages. One of the prominent ones being that Chrome Extensions can install without restarting the browser. The extension starts working immediately.

Much of Chrome's functionality is not exposed to Extensions yet, making extensions such as FlashGot difficult to create due to the lack on an API to monitor downloads. Although with Chrome 4.0 already in beta, and around the corner, we might see extensions take their next step with Chrome 5.0.

An extension from scratch

If you're familiar with web development, you'll be glad to know that a Chrome extension is essentially just JavaScript, HTML and CSS. It's pretty much the same thing as developing web applications, except that you have access to some of the internal functions of Chrome.

Using JavaScript you can manipulate parts of Chrome's functionality, or appearance. Depending on what you want your extension to do, you can listen to events within Chrome such as changing the current tab, or adding a

Polishing your Chrome

Google Chrome finally gets support for extensions; grab a piece of the action!

bookmark, or you can add a button to the toolbar that presents people additional tools.

Since Chrome development is essentially the same as writing functionality for any web page, you can use JavaScript libraries such as jQuery to make your job much easier. In fact, you can even use Google Analytics with your extension to get statistics about its usage!

We're assuming basic level of knowledge about web development using JavaScript, HTML and CSS. A little knowledge of AJAX will also help since you will often need to load external resources.

Getting started

Since Chrome extensions are just text, JavaScript and HTML (and CSS) files, all you really need is a decent text editorsuch as Notepad++.

A Chrome extension will also need to contain an icon for the extension, and any other resources you might use, such as images, or even videos. You'll also need to install at least the latest beta if not the dev version.

Anatomy

A Chrome extension is a signed ZIP file that contains all the files you need. You needn't worry about packing and signing the file, as Chrome will manage this for you. Since Chrome doesn't need to be restarted after adding an extension, you will immediately be able to see your extension, and in fact you can make changes to your extension while Chrome is running, and reload just your extension to see the latest changes instantly!

A `manifest.json` file in the extension is used to provide Chrome basic information about your extension, such

as a name, description, version, permissions, and information about the kind of extension it is, and where the resources Chrome needs to know about are located. As you can tell by the extension, this is a JSON (JavaScript Object Notation) file that uses a JavaScript-like format for storing information.

Other than that, the extension will contain at least one HTML file to contain the code for the extension, and it may contain other resources such as icons, images and JavaScript files.

To begin with, you should create a folder to contain all your extensions files.

manifest

Since the manifest files is just JSON, which is a plain text format, simply create a new text file in your favourite text editor, and save it as `manifest.json` in the directory you created to contain all your extension files.

After creating the file, add the following code to it:

```
{
  "name": "My first Chrome
extension",
  "version": "1.0",
  "description": "Here is a
little something about what my
extension does.",
  "options_
page": "options.html",
  "browser_action": {
    "default_icon": "icon.png",
    "popup": "popup.html"
  },
  "permissions": [
    "http://www.thinkdigit.com/"
  ]
}
```

Besides the obvious name, description, and version entries, we can see the `browser_action` and its sub-fields `default_icon` and `popup` `options_page` and `permissions`.

The `options_page` entry tells Chrome which file should be used for displaying the configuration page for the extensions.